Analytical Chemistry Cumulative Exam (January 26, 2023)
Announced Topic: Vibrational Spectroscopies (Raman and Infrared)

Unless otherwise indicated, please show your work through use of diagrams, explanations and/or calculations in your answers to all of the questions below. If your work is not shown, then this will result in a zero on that problem, even if the answer is correct.

1. **(23 points)** Consider **carbon dioxide** and **water**. Please answer the following questions for ***each*** of these two molecules.

   (a) (4 points) Predict how many normal vibrational modes each species should have.

   (b) (7 points) Diagram each type of vibration using a drawing of the chemical structure and either arrows, "+" sign or "-" sign (or all of these designations) to illustrate each of these types of normal vibration modes. Also provide a name for the vibration (i.e. *stretching vibrations*: symmetric or asymmetric; *bending vibrations*: out-of-plane twisting, out-of-plane wagging, in-plane rocking, in-plane scissoring.)

   (c) (7 points) Explain whether each of these vibrational modes are infrared or Raman active. (Remember to explain what phenomenon makes a vibrational mode infrared active and what phenomenon makes a vibrational mode Raman active).

   (d) (5 points) Just because a vibrational mode is predicted to be active in the infrared or Raman or both, doesn't mean that the vibration produces a strong peak. What are additional considerations that contribute toward producing a strong peak? (Your answer to this question does not have to pertain specifically to water or carbon dioxide. However, you could use those molecules as examples in your discussion, if you'd like.)

2. **(5 points)** Would it be better to use infrared or Raman spectroscopy to study analytes in aqueous solutions? Why? Please include your answers from both #1 (c) and #1 (d) above in your discussion.

3. (12 points) For ***each*** of the molecules and vibrations to consider (a – f), on your answer sheet ***write only one*** of the following options for activity of the vibrations. (You do ***NOT*** need to explain your answers here.) **Select from:**

Only Infrared-active
Only Raman-active
Both infrared-active AND Raman-active
Neither infrared-active nor Raman-active

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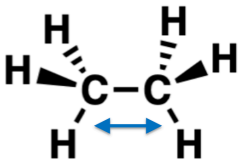
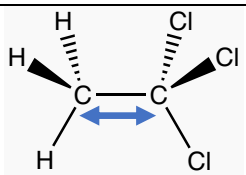
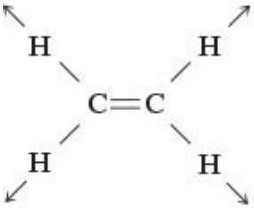
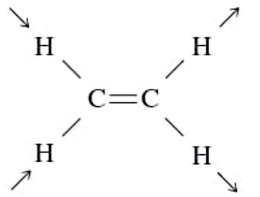
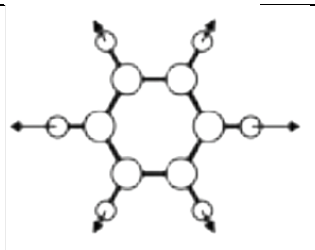
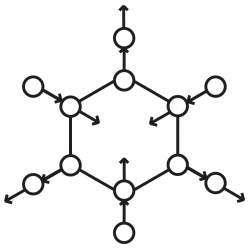
Molecule and the Vibration to Consider	
(a) C-C stretching, Ethane, C_2H_6	
(b) C–C stretching, 1,1,1-trichloroethane, $CH_3–CCl_3$	
(c) C-H stretching, Ethylene, C_2H_4	
(d) C-H stretching, Ethylene, C_2H_4	
(e) C-H stretching, Benzene, C_6H_6 (assume all arrows are of equal length)	
(f) Vibration as depicted by arrows in diagram, Benzene, C_6H_6	

4. (10 points) FT-infrared and the Michelson Interferometer

The figure below shows a schematic of the optics of a single beam FT-infrared spectrometer that includes the Michelson Interferometer. Match the letter of the components (**a – j**) with the name of the component in the list. Only one letter per name of component.

Names of components:

(1) Sample compartment

(2) Beamsplitter

(3) infrared source

(4)infrared transducer

(5) Interferometer

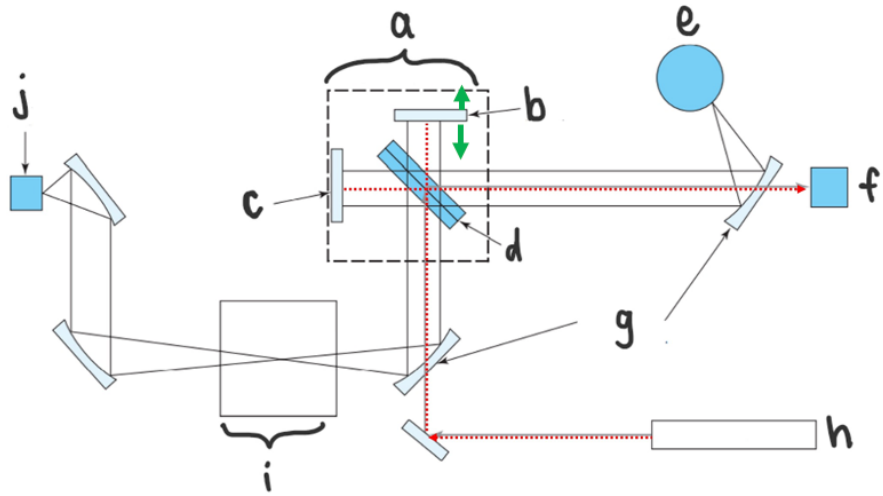
(6) Laser detector

(7) Laser

(8) Mirrors with central hole for laser beam

(9) Movable mirror

(10) Fixed mirror



5. **(8 points)** Why are nearly all infrared spectrometers that are produced today FT-systems instead dispersive instruments? Also, explain what the Jacquinot and the Fellgett advantages are.

6. (8 points) Draw a block diagram for a Raman spectrometer. Be sure to include blocks for the light source (also mention the kind of light source and a possible wavelength), the sample cell, radiation transducer, signal processing and readout system, wavelength selector, as well as a notch filter (and what radiation it filters out).

7. (8 points) Name an EM-radiation source, EM-radiation detector, and examples of suitable materials for sample cells and windows that you would use to perform a mid-infrared FT spectroscopy experiment. Provide explanations for your selections.

8. (8 points) Name an EM-radiation source, EM-radiation detector, and examples of suitable materials for sample cells and windows that you would use to perform a Raman spectroscopy experiment. Provide explanations for your selections.

9. (10 points) Comment on the sensitivity of normal Raman relative to infrared spectroscopy. Name some technological and methodological advances that have made Raman spectroscopy much more feasible and popular than it was a several decades ago?

10. (8 points) Because of the needs of a research project that you are involved in, you decide to build a FTIR instrument based on a Michelson interferometer that will be able to resolve peaks at **1650.000 cm^{-1} and 1650.005 cm^{-1}** in an infrared spectrum. (You can't find any commercially-available instruments that can achieve these specifications that are within your price range, and thus, this is why you want to build one.) However, to keep the price down, you decide to use an off-the-shelf photon transducer (detector) that is a bit slow. Because of the slow response of the transducer, you will have to produce a frequency at this transducer of only **320 Hz** when the light leaving the source is **1650.005 cm^{-1}.**

What are the **mirror speed and the mirror drive length** that you would have to include as part of your design? Would this instrument be practical to build and operate? Why or why not?

Equations, Tables, and Constants for use with Analytical Chemistry Cume

These equations and constants might or might not be helpful for answering questions on the Cume.

c = 2.99792 x 10^8 m/s
h = 6.62608 x 10^{-34} J•s
1 J = 6.2415 x 10^{18} eV
0 °C = 273.15 K
Room temperature = 25 °C = 273.15 + 25 = 298.15 K
Radians = (Angle in degrees)(π)/ (180°)
Refractive index for air = 1.00

$$\nu = \frac{c}{\lambda}$$

$$\frac{1}{\lambda} = \frac{1}{v}\nu = \bar{\nu}$$

$$\eta = \frac{c}{v}$$

$$I_{scattered} \propto \frac{1}{\lambda^4}$$

$$E = \frac{h\,v}{\lambda} = h\nu = hv\bar{\nu}$$

$$E = \frac{h\,c}{\lambda} = h\nu = hc\bar{\nu}$$

$$v_M = \delta/(2\,t)$$

$$f = (2\,v_M)\,\nu\,/\,c$$

$$\Delta\bar{\nu} = \bar{\nu}_2 - \bar{\nu}_1 = 1/\delta$$

$$\delta\bar{\nu}_2 - \delta\bar{\nu}_1 = 1$$

$$\delta = 2(M - F)$$

$$\frac{\delta}{\lambda} = N$$